torch.logaddexp#

https://pytorch.org/docs/stable/generated/torch.logaddexp.html#torch.logadde

Description:

Logarithm of the sum of exponentiations of the inputs. Calculates pointwise log(ex+ey)\log\left(e^x + e^y\right)log(ex+ey). This function is useful in statistics where the calculated probabilities of events may be so small as to exceed the range of normal floating point numbers. In such cases the logarithm of the calculated probability is stored. This function allows adding probabilities stored in such a fashion. This op should be disambiguated with torch.logsumexp() which performs a reduction on a single tensor. input (Tensor) – the input tensor. other (Tensor) – the second input tensor

Function Signature

```
torch.logaddexp(input, other, *, out=None) → Tensor#
Parameters
Keyword Arguments
```

Parameters

Keyword

out (Tensor, optional) – the output tensor.

Code Examples

```
>>> torch.logaddexp(torch.tensor([-1.0]), torch.tensor([-1.0,
-2, -3]))
tensor([-0.3069, -0.6867, -0.8731])
>>> torch.logaddexp(torch.tensor([-100.0, -200, -300]),
torch.tensor([-1.0, -2, -3]))
tensor([-1., -2., -3.])
>>> torch.logaddexp(torch.tensor([1.0, 2000, 30000]),
torch.tensor([-1.0, -2, -3]))
tensor([1.1269e+00, 2.0000e+03, 3.0000e+04])
```

Detailed Documentation

Documentation

Logarithm of the sum of exponentiations of the inputs.

Calculates pointwise log(ex+ey)\log\left(e^x + e^y\right)log(ex+ey). This function is useful in statistics where the calculated probabilities of events may be so small as to exceed the range of normal floating point numbers. In such cases the logarithm of the calculated probability is stored. This function allows adding probabilities stored in such a fashion.

This op should be disambiguated with torch.logsumexp() which performs a reduction on a single tensor.

input (Tensor) – the input tensor.

other (Tensor) – the second input tensor

out (Tensor, optional) – the output tensor.